

from surveillance cameras and identify suspicious activities or potential security threats. With MLOps, these models can be continuously updated and deployed to provide real-time monitoring, enabling law enforcement to respond swiftly to incidents and ensure the safety of residents.

MLOps can also help optimise energy consumption in smart cities. Machine learning algorithms can predict energy demand patterns, enabling cities to allocate resources efficiently and make data-driven decisions for energy distribution. MLOps facilitates the automatic updating of these predictive models, ensuring they remain accurate as energy usage patterns evolve.

It can be leveraged for environmental monitoring and sustainability initiatives. By analysing data from various sensors placed throughout the city, machine learning models can provide insights into air quality, water usage, and noise pollution levels. This data-driven approach empowers policymakers to implement targeted strategies for environmental conservation, and create greener and more sustainable urban spaces.

The use of MLOps in smart cities empowers urban planners, administrators, and citizens to make informed decisions based on real-time data and predictive analytics. It enhances the efficiency of city services, optimises resource allocation, and fosters a more livable, safe, and sustainable environment for the growing urban population. As smart cities continue to evolve, MLOps will undoubtedly play a central role in shaping the future of urban living.

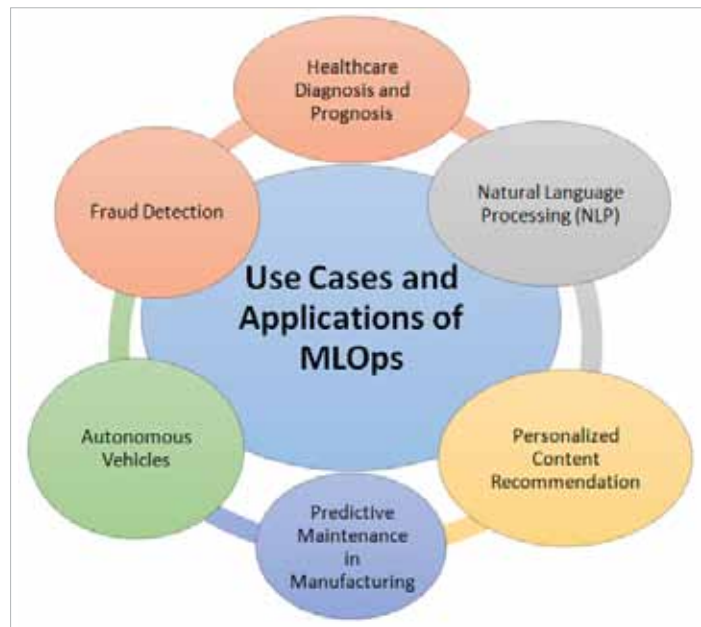


Figure 2: Use cases and applications of MLOps

## Real world use cases and applications of MLOps

The practical application of MLOps across various industries demonstrates its transformative impact on diverse fields, from e-commerce and manufacturing to healthcare and finance. These real-world use cases exemplify how MLOps optimises processes, enhances decision-making, and ushers in a new era of efficiency and innovation.

### **Automated image classification**

**in e-commerce:** An e-commerce company utilises ML models to classify products automatically. MLOps helps in continuous training and deployment of these models as new product images are added or existing ones change. By automating the model deployment process, the company can maintain the accuracy of its classification system, ensuring customers receive relevant search results.

**Predictive maintenance in manufacturing:** A manufacturing plant employs ML models to predict machinery failures and schedule preventive maintenance. MLOps

enables the seamless integration of these predictive models into their operational systems. This ensures that maintenance schedules are automatically updated based on real-time data, reducing downtime and increasing overall efficiency.

**Fraud detection in finance:** A financial institution uses ML models to detect fraudulent transactions. MLOps helps in automating the training and deployment of these models, ensuring

that the fraud detection system remains up-to-date with the latest patterns and trends in fraudulent activities.

**Healthcare diagnosis and prognosis:** Hospitals implement ML models for medical image analysis, patient diagnosis, and treatment prognosis. MLOps ensures the continuous improvement and deployment of these models, making sure that medical professionals have access to the most accurate and reliable tools to aid in patient care.

**Natural language processing (NLP) for customer support:** Companies deploy NLP models to analyse and categorise customer support tickets. MLOps streamlines the process of updating and deploying these models as new customer queries and support ticket data are generated.

**Autonomous vehicles:** Companies working on autonomous vehicles require continuous development and deployment of ML models for perception, decision-making, and control systems. MLOps helps ensure that the most recent and optimised models are used, improving the safety and performance of autonomous vehicles.